INDIVIDUAL THESIS

AI-BASED VIRTUAL TRY-ON AND SIZE RECOMMENDATION SYSTEM

An Individual Component of the Smart Omni-Retail System

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Group ID: R26-IT-060

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Selecting apparel online remains difficult when consumers cannot physically assess garment appearance or interpret brand-dependent size labels. This individual research component combines body measurement prediction, standard and brand-specific sizing, generative virtual try-on, and retrieval-augmented styling assistance. A multi-output Random Forest regressor estimates shoulder width, waist circumference, and leg length from height, weight, gender, and a derived body mass index. A rule-based standard-size mapping is followed by a Random Forest classifier for brand-specific size prediction. The system integrates Google Cloud Vertex AI virtual-try-on-001 for customer–garment image generation, Flutter for customer interaction, FastAPI for services, Firebase Firestore for profiles, and ChromaDB with a Gemini-based conversational stylist. The supplied research paper reports regression mean absolute errors of 1.12 cm, 1.45 cm, and 1.28 cm, with corresponding R^2 values of 0.914, 0.897, and 0.921, and brand-size classification accuracy of 88.4%. These are reported findings, not independently reproduced results in this draft. The work provides an integrated digital fitting workflow, while limitations include dataset representativeness, photo artefacts, unseen brands, and dependence on third-party generative services. Further independent evaluation and implementation evidence are required before final submission.

Keywords: virtual try-on; Random Forest; apparel sizing; retrieval-augmented generation; personalized fashion

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
ERP	Enterprise Resource Planning
FR	Functional Requirement
IEEE	Institute of Electrical and Electronics Engineers
IP	Intellectual Property
LLM	Large Language Model
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
NFR	Non-Functional Requirement
NoSQL	Not only Structured Query Language
PDPA	Personal Data Protection Act
POS	Point of Sale
PP1, PP2, PP3	Progress Presentations 1, 2 and 3
R^2	Coefficient of Determination
RMSE	Root Mean Squared Error
ROI	Return on Investment
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
XAI	Explainable Artificial Intelligence

Abbreviation	Meaning
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall Project Context

Fashion retail is increasingly supported by digital tools that help customers discover products and help retailers improve the shopping experience. Online apparel shopping presents a specific challenge: customers cannot physically try on a garment before purchasing it, and clothing sizes can differ between brands and garment types. Visual product images alone may not communicate how a garment could appear on an individual customer, while a generic size chart may not reflect a customer's body proportions.

The wider AI-Driven Smart Fashion Retail Recommendation System is described in the supplied research paper as an integrated platform containing four components: inventory and stock-flow optimization, personalized marketing intelligence, virtual try-on and size recommendation, and GPS-based customer behaviour tracking with staff assistance. The components use shared backend services and Firebase Firestore. This thesis focuses only on the virtual try-on and size recommendation component; other components are discussed only where needed to explain the wider system and integration.

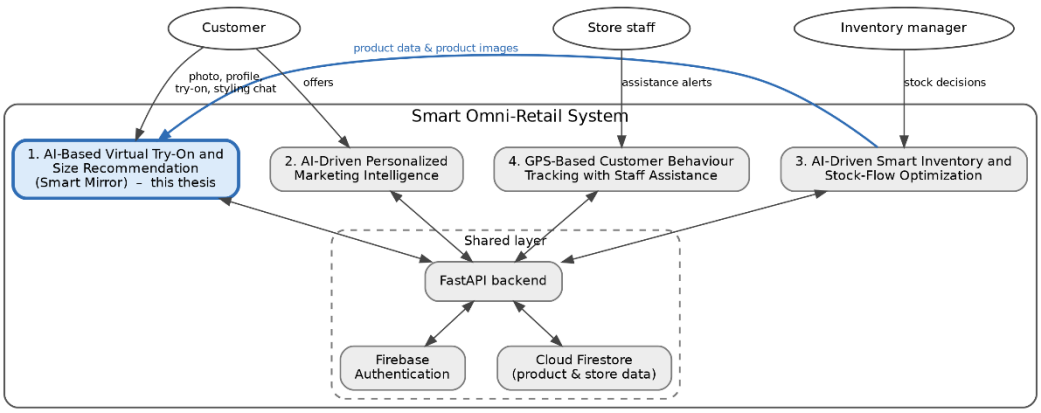


Figure 1.1: Overall Smart Omni-Retail System context

Table 1.1: Components of the Smart Omni-Retail System

No.	Component	Member	Student ID
1	AI-Based Virtual Try-On and size Recommendation System	Silva T. A	IT22284952
2	AI-Driven Personalized Marketing Intelligence	WIJETUNGA J A R K	IT22146588
3	AI-Driven Smart Inventory and Stock-Flow Optimization	W.M Methsara	IT22244598
4	GPS-Based Customer Behaviour Tracking with Staff Assistance	Perera L. A.D. A. S	IT22243812

1.2 Individual Component Context

The AI-Based Virtual Try-On and Size Recommendation System is designed to help customers select suitable apparel through an intelligent digital fitting-room experience. Its main functions include predicting body measurements using a Random Forest Regressor, recommending a standard clothing size based on customer attributes, predicting brand-specific sizes using a Random Forest Classifier, generating realistic virtual try-on images using Google Cloud Vertex AI, and providing personalized styling and fit recommendations through a Retrieval-Augmented Generation (RAG)-based AI stylist.

The system uses height, weight, gender, and calculated BMI to predict shoulder width, waist circumference, and leg length. It also integrates product details, descriptions, and customer reviews into a ChromaDB vector database to retrieve relevant information and generate personalized styling advice using Gemini AI. The component is implemented using Flutter, FastAPI, Firebase Firestore, and machine-learning technologies to improve customers' online apparel shopping experience.

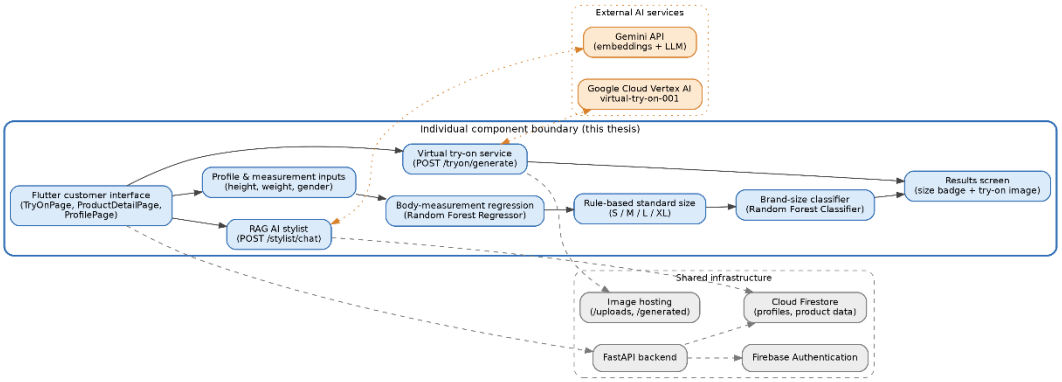


Figure 1.2: Individual component scope within the overall solution

1.3 Problem Statement

Online clothing shopping presents significant challenges for customers in selecting suitable garment sizes and understanding how clothes will look on their bodies before purchasing. Customers mainly depend on product images, descriptions, and standard size charts, which may not accurately represent individual body proportions or differences in sizing standards across clothing brands. These limitations can lead to incorrect size selections, poor fitting experiences, customer dissatisfaction, and increased product returns. Although existing virtual try-on technologies provide digital garment visualization, many approaches do not fully integrate personalized body measurement prediction with brand-specific size recommendations [4]. Furthermore, some existing solutions require manual body measurements or specialized scanning equipment, making the fitting process less convenient for customers.

Therefore, this research addresses the need for an intelligent digital fitting-room system that integrates machine-learning-based body measurement estimation, brand-specific size prediction, AI-generated virtual try-on, and personalized styling recommendations. The proposed system aims to improve the online apparel shopping experience by helping customers visualize selected garments and identify suitable clothing sizes using basic personal information without requiring specialized body-scanning equipment.

The main research problem is: How can AI-based virtual try-on and machine-learning-based size recommendation techniques be integrated to provide personalized garment visualization and accurate brand-specific size recommendations, enabling customers to make more informed clothing purchasing decisions?

1.4 Research Gap

Research on virtual try-on and fashion recommendation systems has advanced significantly with the development of artificial intelligence, machine learning, and computer vision technologies. Image-based virtual try-on approaches such as VITON [4] and characteristic-preserving virtual try-on networks [11] focus on generating realistic garment visualizations while preserving clothing textures and appearance. Similarly, virtual dressing-room systems aim to improve online clothing selection through digital fitting experiences [26]. However, in the literature reviewed for this research, virtual garment visualization and personalized size recommendation are often addressed as separate problems. Existing approaches primarily focus on how garments appear on customers rather than integrating body measurement estimation, standard size prediction, and brand-specific size recommendations into a single customer-facing system.

A second research gap concerns personalized sizing and styling assistance. Clothing sizes vary across brands, making standard size charts less reliable for customers with different body proportions. Machine-learning algorithms such as Random Forest [10] provide opportunities to predict body measurements and recommend suitable clothing sizes using basic customer information. Furthermore, visually aware fashion recommendation systems [15] and large language models [16], [18] support personalized fashion recommendations. However, these approaches do not necessarily combine predicted body measurements, brand-specific sizing information, product descriptions, and customer reviews to provide product-grounded fitting and styling advice. Without integrating these features, customers may receive garment visualizations or general recommendations without sufficient information to determine whether a selected product is suitable for their body measurements and preferences.

The research gap addressed in this study is therefore the limited integration of intelligent body measurement prediction, brand-specific size recommendation, AI-generated virtual try-on, and personalized styling assistance within a unified digital fitting-room system. The proposed component aims to address this gap by combining Random Forest-based measurement estimation and size classification [10], Google Cloud Vertex AI for virtual garment visualization [17], and Retrieval-Augmented Generation (RAG) for product-grounded styling recommendations. This integration is intended to help customers visualize selected garments, identify suitable clothing sizes, and make more informed purchasing decisions without requiring specialized body-scanning equipment. The identified gap is based on the literature reviewed in this research and does not imply that no existing commercial system provides similar functionality.

1.5 Aim and Objectives

The aim of this research is to design, implement and evaluate an integrated AI-Based Virtual Try-On and Size Recommendation System for online fashion retail that predicts customer body measurements, recommends suitable standard and brand-specific clothing sizes, generates realistic virtual garment visualizations, and provides personalized styling and fit recommendations through retrieval-augmented generation.

The objectives derived from this aim are listed in Table 1.2. They are referred to as O1 to O8 in later chapters Section 9.2 reports how far each one was achieved.

Table 1.2: Research objectives

ID	Objective	Addressed in
O1	Prepare and preprocess customer body measurement and clothing size datasets, including height, weight, gender, BMI, body measurements and brand-specific sizing information.	Chapter 3
O2	Develop and evaluate a Random Forest Regression model to predict shoulder width, waist circumference and leg length using basic customer attributes.	Chapters 3, 5, 6
O3	Implement a standard clothing size recommendation mechanism using customer height, weight and predicted body measurements	Chapters 4, 5
O4	Develop and evaluate a Random Forest Classification model to recommend brand-specific clothing sizes based on standard size, clothing brand and product category.	Chapters 3, 5, 6
O5	Integrate Google Cloud Vertex AI to generate realistic virtual try-on images using customer photographs and selected garment images.	Chapters 4, 5, 6
O6	Develop a RAG-based virtual stylist using ChromaDB and Gemini AI to provide personalized styling and fit advice based on product information, customer reviews and predicted measurements.	Chapters 4, 5, 6
O7	Develop a customer-friendly Flutter mobile interface integrated with FastAPI and Firebase to support virtual try-on, size recommendations and personalized styling assistance.	Chapters 4, 5
O8	Evaluate the component through machine-learning performance evaluation, functional testing and integration testing with other components of the AI-driven smart retail system.	Chapters 5, 6

1.6 Research Questions or Hypotheses

Four research questions guide this study. They are presented in Table 1.3 together with the evidence used to answer each question. These are formulated as research questions rather than formal statistical hypotheses because the evaluation focuses on machine-learning performance metrics, virtual try-on image generation, personalized recommendation outputs, and functional testing. No formal statistical hypothesis testing is assumed.

Table 1.3: Research questions

ID	Research question	Evidence
RQ1	How accurately can a Random Forest Regression model predict shoulder width, waist circumference and leg length using customer height, weight, gender and BMI?	MAE, RMSE and R^2
RQ2	How effectively can standard size recommendations and a Random Forest Classification model provide suitable brand-specific clothing sizes?	Standard size rule testing, accuracy, precision, recall and F1-score
RQ3	Can Google Cloud Vertex AI generate virtual try-on images that realistically visualize selected garments on customer photographs?	Generated virtual try-on images, visual comparisons and functional testing
RQ4	Can a RAG-based virtual stylist provide personalized styling and fit recommendations using product information, customer reviews and predicted body measurements?	ChromaDB retrieval results, AI-generated recommendations and integration testing

1.7 Scope, Limitations and Assumptions

The component covers AI-based body measurement prediction, standard and brand-specific clothing size recommendations, virtual garment visualization, and personalized styling assistance through Retrieval-Augmented Generation (RAG). It allows customers to enter basic body information, receive predicted measurements, identify suitable clothing sizes, and visualize selected garments using their photographs. The system uses Random Forest models for measurement and size prediction, Google Cloud Vertex AI for virtual try-on, and ChromaDB with Gemini AI for styling recommendations. It is developed using a Flutter application supported by FastAPI and Firebase.

Several limitations apply. Body measurements are estimated from height, weight, gender, and BMI rather than obtained through direct physical measurement or 3D body scanning. Therefore, predictions may not accurately represent every customer's body proportions. Brand-specific size recommendations depend on the available training data and may be less reliable for unfamiliar brands or clothing categories. Virtual try-on images are AI-generated visualizations and may not accurately represent actual garment fit, fabric behaviour, or physical appearance. The virtual try-on and AI stylist features also depend on external AI services with possible availability, latency, and usage limitations. The system has not been validated through large-scale real-world customer trials, and its impact on product returns or customer satisfaction has not been measured.

The following assumptions are made. Customers provide accurate height, weight, gender, and suitable photographs. Product images and brand information are assumed to be correctly provided. The training datasets are assumed to contain representative body measurements and clothing size information. Product descriptions and customer reviews stored in ChromaDB are treated as reliable sources for generating styling recommendations. Generated virtual try-on images are intended to support purchasing decisions rather than guarantee exact physical garment fit.

1.8 Individual Contribution

The work presented in this thesis represents my individual contribution to the group research project, and I have clearly distinguished my responsibilities from the shared work and contributions of other group members. I developed the AI-Based Virtual Try-On and Size Recommendation System, including the Random Forest Regression model for predicting shoulder width, waist circumference, and leg length, the standard clothing size recommendation mechanism, and the Random Forest Classification model for predicting brand-specific clothing sizes. I also implemented the virtual try-on functionality using Google Cloud Vertex AI and the RAG-based virtual stylist using ChromaDB and Gemini AI. My contribution further includes the FastAPI backend services, relevant Flutter application interfaces, customer profile handling, and functional testing of these features.

Some infrastructure and resources were shared among group members. Firebase Authentication, Firestore, and product information were used as common resources to support integration across the smart retail platform. My component focuses specifically on customer-facing virtual try-on, personalized size recommendations, and AI-based styling assistance. I did not develop the inventory forecasting and stock management component, personalized marketing intelligence component, or GPS-based customer behaviour tracking and staff assistance component. My work was integrated with the shared retail platform to provide a unified customer shopping experience.

1.9 Thesis Structure

The rest of the thesis is organized as follows. Chapter 2 reviews the literature on AI-based virtual try-on, body measurement prediction, clothing size recommendation, machine learning, and retrieval-augmented generation, and presents the conceptual framework. Chapter 3 describes the research methodology, including dataset preparation, Random Forest model development, virtual try-on procedures, evaluation metrics, and ethical considerations. Chapter 4 presents the system requirements, architecture, and detailed design of the proposed solution. Chapter 5 explains the implementation, integration, and testing of the virtual try-on, size recommendation,

and AI stylist functionalities. Chapter 6 presents the machine-learning evaluation results, virtual try-on outputs, and available validation evidence. Chapter 7 discusses the research findings, comparisons with existing approaches, and system limitations. Chapter 8 examines commercialisation opportunities and sustainability considerations, while Chapter 9 concludes the research and outlines future improvements. The references, glossary, and appendices follow, including supporting technical documentation and the required disclosure of generative AI use.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This chapter reviews research in five areas directly related to the component: AI-based virtual try-on, body measurement prediction, clothing size recommendation, machine-learning algorithms, and retrieval-augmented generation for personalized styling assistance. Cloud-based AI services and database technologies are briefly discussed where they influence system design.

The review was targeted rather than exhaustive. Priority was given to relevant journal papers, conference publications, and technical documentation published between 2020 and 2026, while older foundational studies were retained. Important references include VITON [4], characteristic-preserving virtual try-on [11], Random Forest [10], and virtual dressing-room research [26].

2.2 Domain and Theoretical Foundations

2.2.1 AI-Based Virtual Try-On

Virtual try-on technology allows customers to visualize selected garments on their photographs without physically wearing them. Image-based approaches use computer vision and generative models to combine customer and clothing images. Han et al. introduced VITON for realistic garment visualization [4], while Wang et al. developed a characteristic-preserving approach to maintain important clothing details [11]. Virtual dressing-room systems have also been investigated to support online clothing selection [26]. The proposed component uses Google Cloud Vertex AI to generate virtual try-on images [17]. However, generated images do not guarantee accurate physical garment fit.

2.2.2 Body Measurement Prediction Using Random Forest

Body measurement prediction estimates physical dimensions from available customer attributes. Random Forest combines multiple decision trees to model relationships between input features and numerical outputs [10]. It is suitable for structured data and can capture nonlinear relationships. The proposed component uses a Random Forest Regressor to predict shoulder width, waist circumference, and leg length from height, weight, gender, and BMI. GridSearchCV with three-fold cross-validation is used for model optimization. Prediction accuracy depends on training data quality and the variation in customer body proportions.

2.2.3 Standard and Brand-Specific Size Recommendation

Clothing size recommendation helps customers select suitable garments based on their physical characteristics. However, standard size labels such as S, M, L, and XL may differ across clothing brands. Personalized recommendation research highlights the importance of considering customer and product information [1]. The proposed system first determines a standard clothing size using height- and weight-based rules. A Random Forest Classifier then predicts a brand-specific size using the standard size and categorical product attributes [10]. This approach aims to reduce uncertainty caused by inconsistent clothing size standards.

2.2.4 Machine-Learning Evaluation Metrics

Regression and classification models require different evaluation measures. Mean Absolute Error (MAE) measures the average difference between predicted and actual body measurements, while Root Mean Squared Error (RMSE) gives greater weight to larger errors. The coefficient of determination (R^2) measures how much variation is explained by the regression model. For brand-specific size classification, accuracy measures correct predictions, while precision, recall, and F1-score provide additional information about classification performance. These metrics help evaluate model reliability, although results depend on the dataset and testing procedure.

2.2.5 Personalized Styling and Retrieval-Augmented Generation

Personalized fashion recommendation systems use customer preferences and product information to suggest suitable clothing [1], [15]. Large language models can generate conversational styling advice, but their responses may contain unsupported information [16]. Retrieval-Augmented Generation (RAG) helps address this issue by retrieving relevant product information before generating responses. The proposed component uses ChromaDB to retrieve product descriptions, metadata, and customer reviews, while Gemini AI generates personalized styling and fit recommendations [18]. Predicted customer measurements are also included to provide additional context. However, recommendation quality depends on the accuracy of retrieved information and generated responses.

2.3 Related Research

2.3.1 Image-Based Virtual Try-On

Han et al. introduced VITON, an image-based virtual try-on network that generates realistic garment visualizations using customer and clothing images [4]. Wang et al. proposed a characteristic-preserving approach to improve garment alignment and maintain clothing details [11]. Weerasinghe et al. explored virtual dressing-room technology for online clothing selection [26]. These studies mainly focus on garment visualization. The present component extends this approach by integrating virtual try-on with body measurement prediction and clothing size recommendations.

2.3.2 Machine Learning for Body Measurement Prediction

Breiman introduced Random Forest, an ensemble algorithm that combines decision trees for regression and classification [10]. Its ability to model nonlinear relationships makes it suitable for predicting body measurements from customer information. [CITATION REQUIRED: a study applying Random Forest to body measurement estimation.] The proposed component uses Random Forest Regression to estimate shoulder width, waist circumference, and leg length from height, weight, gender, and BMI. Unlike body-scanning approaches, this method requires only basic customer information.

2.3.3 Personalized Clothing Size Recommendation

Clothing size differences across brands create uncertainty in online shopping. Personalized recommendation research highlights the importance of customer and product attributes when generating recommendations [1]. Random Forest Classification can identify relationships between structured attributes and categorical size labels [10]. The proposed component combines rule-based standard sizing with machine-learning-based brand-specific size classification to provide more relevant recommendations.

2.3.4 Personalized Fashion Recommendation

Kang et al. investigated visually aware fashion recommendation using garment images and generative models to improve personalized product suggestions [15]. Such approaches focus on visual preferences and product characteristics but do not necessarily consider individual body measurements. The present component combines predicted measurements with product descriptions, brand information, and customer reviews to provide personalized styling advice. This supports recommendations based on both customer characteristics and available product information.

2.3.5 Retrieval-Augmented Generation for Virtual Styling

Retrieval-Augmented Generation (RAG) combines information retrieval with language models to generate responses grounded in external knowledge. This approach helps reduce unsupported recommendations, although it cannot eliminate hallucinations. [CITATION REQUIRED: a foundational RAG research paper.] The proposed component uses ChromaDB to retrieve relevant product information and Gemini AI to generate personalized styling recommendations. Customer measurements and retrieved product details provide additional context for the responses. This approach connects conversational assistance with the virtual try-on and size recommendation functionalities.

2.4 Existing Systems, Products and Techniques

Existing virtual try-on and clothing size recommendation solutions can be grouped into four broad approaches. The first is traditional online shopping, where customers

select clothing using product photographs and standard size charts. The second is image-based virtual try-on technology, such as VITON, which generates garment visualizations using customer and clothing images [4]. The third is virtual dressing-room systems that support digital clothing selection [26]. The fourth is machine-learning-based recommendation techniques that use customer attributes and product information to suggest suitable clothing [10], [15].

These approaches offer different capabilities, but they do not necessarily combine body measurement prediction, brand-specific sizing, virtual try-on, and personalized styling assistance within one solution. The strengths and limitations identified from the reviewed literature are summarized in Table 2.1. The comparison is the author's synthesis and should be verified against the cited sources.

Table 2.1: Indicative comparison of approaches (author's synthesis)

Approach	Body Measurement Prediction	Brand-Specific Sizing	Virtual Try-On	AI Styling Assistance
Traditional online shopping	No	Size charts only	No	Usually not included
Image-based virtual try-on research	Not the main focus	Not the main focus	Yes	Not the main focus
Virtual dressing-room research	Varies	Varies	Yes	Varies
Personalized fashion recommendation research	Not the main focus	Not the main focus	Not necessarily	Product recommendations
This component	Random Forest Regression	Random Forest Regression	Google Cloud Vertex AI	RAG with ChromaDB and Gemini AI

2.5 Gap Synthesis and Conceptual Framework

Three observations follow from the review. First, image-based virtual try-on research focuses mainly on garment visualization, with limited integration of personalized body

measurement prediction. Second, clothing size recommendation approaches address sizing uncertainty, but standard sizes may not account for differences between brands. Third, personalized fashion recommendation systems focus on product preferences and visual characteristics, while the integration of predicted body measurements with retrieval-augmented styling advice remains an area for further investigation.

These observations shaped the conceptual framework in Figure 2.1. Customer information, including height, weight, and gender, is provided to a Random Forest Regression model to predict body measurements. A rule-based mechanism determines the standard clothing size, while a Random Forest Classifier generates brand-specific size recommendations. Customer photographs and garment images are processed through Google Cloud Vertex AI to generate virtual try-on images. A RAG-based virtual stylist retrieves product information from ChromaDB and uses Gemini AI to provide personalized styling recommendations. The Flutter application integrates these features through FastAPI services, with Firebase supporting customer data management.

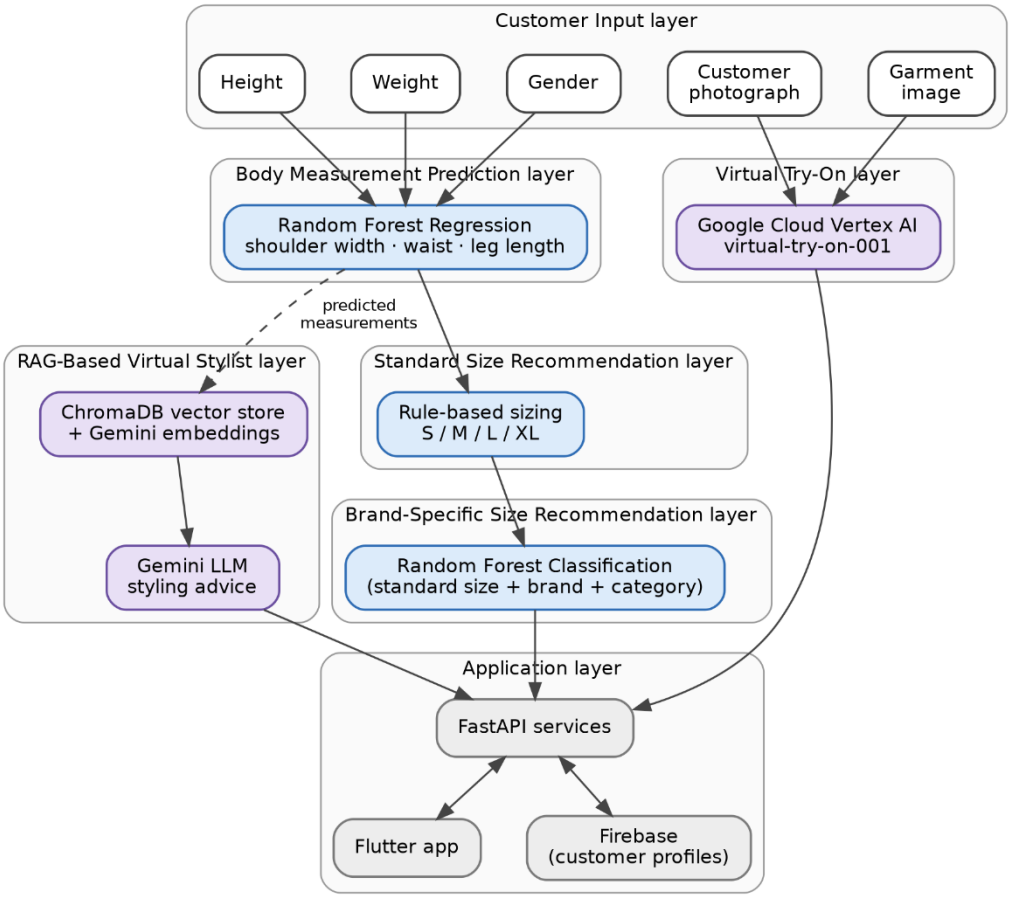


Figure 2.1: Conceptual framework for the AI-Based Virtual Try-On and Size Recommendation System.

3 RESEARCH METHODOLOGY

3.1 Research Design

The project is best described as a design-and-evaluate study: a software component is designed and implemented, and its predictive and technical behaviour is evaluated using evidence appropriate to each function. The measurement regressor is evaluated as a regression model; the brand-size classifier is evaluated as a classification model; the virtual try-on service is assessed separately for successful processing and visual quality; and the integrated application is assessed through functional and end-to-end testing.

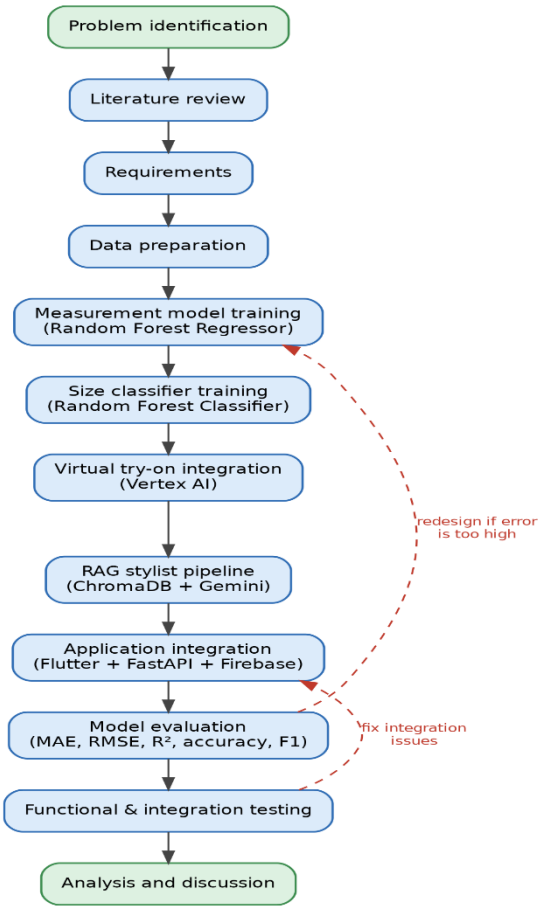


Figure 3.1: Research methodology workflow

3.2 Data, Participants or Study Materials

3.2.1 Data source and its limits

The supplied paper reports model outputs but does not fully specify the dataset source, record count, date range, collection procedure, or data licensing. These must be supplied from the project artefacts. State clearly whether the data is public, collected, synthetic, anonymized, or prepared for the project. If data is synthetic, do not describe it as real customer measurements.

3.2.2 Dataset attributes

Table 3.1: Dataset Details and Information to Be Confirmed

Dataset / data group	Known from current paper	Must be confirmed
Measurement dataset	Inputs include height, weight, gender, BMI; targets include shoulder width, waist circumference, leg length	Source, number of rows, units, missing values, distributions, inclusion criteria
Brand-size dataset	Classifier maps standard size to brand-specific size using categorical product attributes	Brands, labels, features, number of samples per class, class imbalance
Person images	Used as input for virtual try-on	Source/consent, resolution, file formats, pose/garment constraints, storage and deletion
Garment images	Selected garment image used for visualization	Source, rights, catalogue coverage, image requirements
RAG knowledge collection (if used)	Product metadata, descriptions, customer reviews; embeddings in ChromaDB	Number of documents, source, chunking, metadata fields, update process

3.2.3 Data preparation and preprocessing

Describe duplicate removal, missing-value handling, categorical encoding, unit normalization, outlier checks, feature derivation, and train/test separation. BMI should be calculated consistently using weight in kilograms and height in metres: BMI = weight (kg) / height (m)^2. Confirm whether BMI is computed in code or supplied as a dataset column. Any transformation must be fitted on training data only where relevant to avoid leakage.

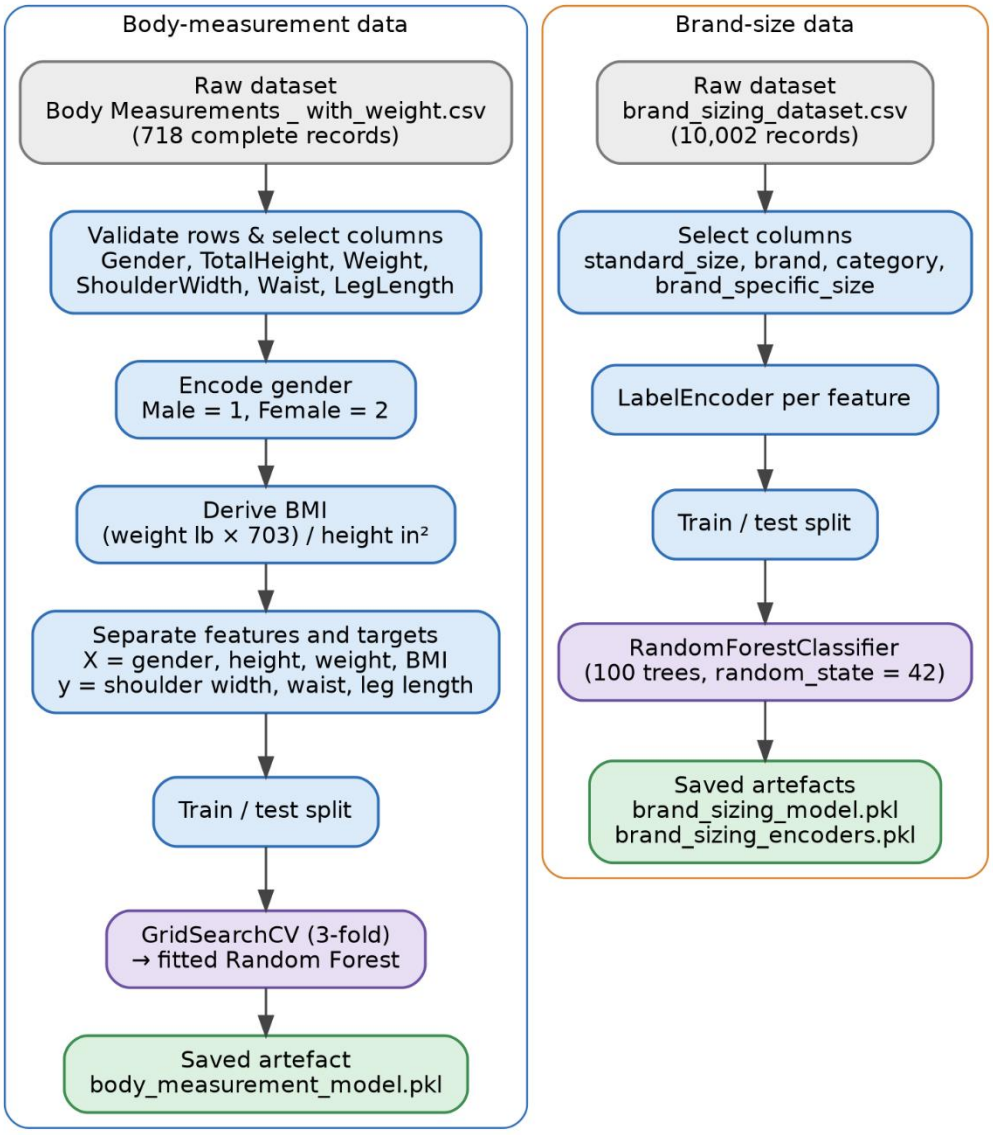


Figure 3.2: Data preparation and feature-engineering pipeline

3.3 Experimental or Study Procedure

3.3.1 Body-measurement regression

The project paper describes a Random Forest Regressor configured for multi-output prediction. The features are height, weight, gender, and BMI; the target vector contains shoulder width, waist circumference, and leg length. GridSearchCV with three-fold cross-validation is reported for model optimization. The final thesis must state the search space, scoring method, random seed, split procedure, selected hyperparameters, software versions, and whether all three targets were trained jointly or through separate models.

3.3.2 Brand-specific size classification

The paper reports a 100-tree Random Forest Classifier that maps a predicted standard size to a brand-specific size using categorical product attributes. Describe the exact input features, target labels, encoding, class balancing, train/test split, and whether the classifier receives the regression model's predicted measurements or a derived standard-size label. This distinction matters because using ground-truth sizes during evaluation could overstate end-to-end performance.

3.3.3 Virtual try-on procedure

Document the exact provider/model identifier, API request format, image preparation, prompt or parameters, maximum image size, output format, timeout, retry logic, failure response, and image storage. The current paper names Google Cloud Vertex AI's virtual-try-on-001, while the user's presentation description calls the service Gemini-based. Confirm the actual endpoint from implementation evidence before finalizing.

3.3.4 RAG stylist procedure (if included)

The supplied paper describes product metadata, descriptions, and customer reviews being embedded with all-MiniLM-L6-v2, stored in ChromaDB, and retrieved using LangChain vector similarity. The retrieved context is passed to a language model to produce product-grounded recommendations. Add the actual chunk size, overlap, top-k value, filters, model identifier, prompt, source display, fallback, and evaluation method. If these settings are unknown, mark them as unreported rather than inventing values.

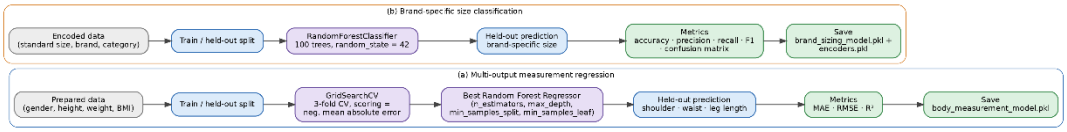


Figure 3.3: Machine-learning training and evaluation workflow

3.4 Measures and Evaluation Metrics

Table 3.2: Model Evaluation Metrics and Reporting Requirements

Task	Metric	Interpretation / reporting requirement
Measurement regression	MAE	Mean absolute error in the target's physical unit; report separately for each measurement.
Measurement regression	RMSE	Penalizes larger errors more strongly; report units and test set.
Measurement regression	R^2	Variance explained relative to a mean-prediction baseline; not a direct measure of centimetre error.
Brand-size classification	Accuracy	Fraction of correct predictions; can hide minority-class errors.
Brand-size classification	Precision / recall / F1	Report per-class and macro/weighted averages as appropriate.
Brand-size classification	Confusion matrix	Shows which size classes are confused.
Virtual try-on	Functional success rate and response time	Report request counts, failure cases, and timing method.
Virtual try-on	Structured visual quality assessment	Use defined criteria or a documented human-rating protocol; do not infer from regression metrics.
RAG stylist	Groundedness / retrieval relevance	Use a defined test set and rubric; include unsupported-claim checks if assessed.

3.5 Data Analysis

3.6 Ethics, Privacy, Security and Responsible AI

The component involves customer body information and photographs, making privacy an important consideration. The body measurement and brand-specific sizing models were developed using prepared datasets rather than a documented study involving real customers. Customer height, weight, gender, predicted measurements, and uploaded photographs are processed to provide personalized recommendations and virtual try-on results. If real customer data is used, its collection, storage, and processing must comply with Sri Lanka's Personal Data Protection Act, No. 9 of 2022.

On security, user authentication is supported by Firebase Authentication, while Firestore stores customer profile information. FastAPI handles requests to the machine-learning models and external AI services. Uploaded photographs are processed through Google Cloud Vertex AI for virtual try-on generation. API keys and customer information should be protected against unauthorized access.

Responsible AI considerations include measurement accuracy, recommendation reliability, and generated-image limitations. Predicted measurements may contain errors, and virtual try-on images cannot guarantee actual garment fit. The RAG-based stylist uses retrieved product information to support recommendations, but generated advice may still be inaccurate. The system is intended to assist customers rather than replace their purchasing decisions. Generative AI use in thesis preparation is disclosed in Appendix B.

3.7 Reproducibility and Research Artefacts

Record source-code revision, environment versions, dataset version, preprocessing steps, random seeds, selected hyperparameters, saved model identifiers, API/model endpoint, and test records. Keep secrets and personal images out of public

repositories. Where data cannot be redistributed, document its provenance and provide a schema or synthetic example that does not disclose personal information.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Functional requirements

Table 4.1: Functional requirements

ID	Requirement	Verification method
FR-01	The customer can provide or update the supported profile attributes.	UI test with valid and invalid values
FR-02	The system validates profile inputs and units before prediction.	Boundary and validation tests
FR-03	The system predicts the supported body measurements.	Compare API output schema and saved model prediction
FR-04	The system returns a size recommendation for a supported brand.	Test known supported brands and size classes
FR-05	The customer can select/upload a garment image and provide a person image.	Image input tests
FR-06	The system requests a virtual try-on result and displays the output or a clear error.	Success, timeout, and provider-error tests
FR-07	The system presents the predicted size separately from the visual preview.	UI inspection
FR-08	If implemented, the stylist retrieves product context before generating advice.	Retrieval and grounding tests
FR-09	The system handles missing inputs, unsupported brands, and service failures safely.	Negative testing

4.1.2 Non-functional requirements

Table 4.2: Non-functional requirements

ID	Requirement area	Design consideration / target
NFR-01	Usability	Clear instructions for images and profile inputs; explain that preview is approximate.
NFR-02	Performance	Measure actual prediction and image-generation response times; do not set unsupported claims.
NFR-03	Reliability	Timeouts, error messages, retry limits, and fallback behaviour.
NFR-04	Security	Authentication/authorization where required, secure API key handling, encrypted transport.
NFR-05	Privacy	Purpose limitation, minimal retention, deletion process, access control.
NFR-06	Maintainability	Separate model, API, image-generation, and presentation responsibilities.
NFR-07	Compatibility	Document supported image formats, sizes, devices, and browsers.

4.2 Component Architecture Within the Complete Solution

At component level, the frontend gathers profile attributes and images, sends requests to the backend, and displays measurements, size advice, and the generated visualization. The backend validates inputs, invokes the measurement and size models, calls the virtual try-on service, and returns structured results. Firebase Firestore is identified in the research paper as the shared data store for customer profile information and predicted measurements. Product metadata and images support the optional RAG pipeline.

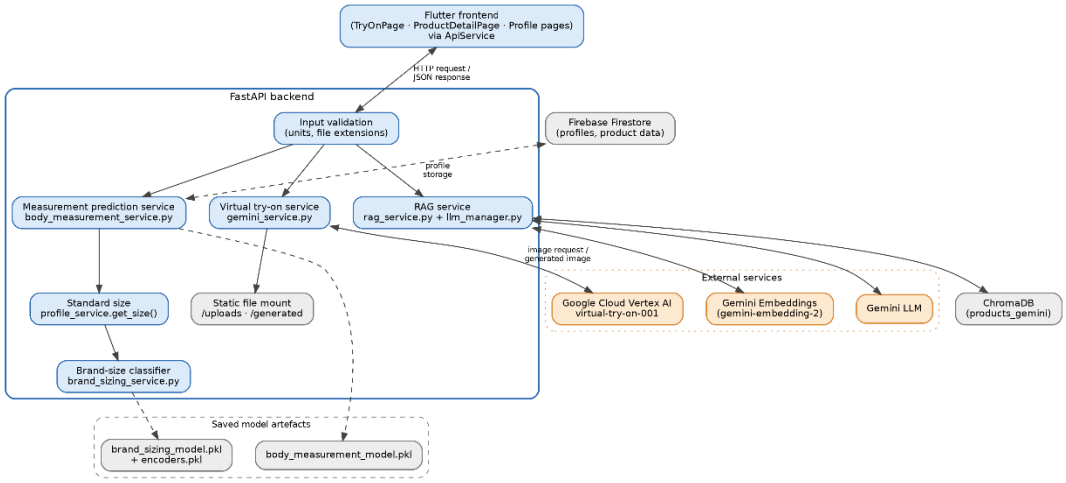


Figure 4.1: Overall architecture of the individual component

4.3 Design Alternatives and Decisions

Table 4.3: Design Alternatives and Decisions

Decision	Alternative(s)	Rationale to document	Evidence needed
Measurement prediction	Rule-based size chart; linear model; Random Forest	Tabular nonlinear model selected in the project paper; confirm actual comparison performed.	Model configuration and evaluation
Size mapping	Fixed size chart; classification model	Brand-specific classifier is reported; explain available labelled data.	Brand/label coverage and metrics
Virtual try-on	Manual photo editing; 3D avatar/scan; managed AI API	Project targets an image-based API workflow without dedicated scanning.	Actual API endpoint and input/output examples
Stylist advice	Static rules; direct LLM prompt; RAG	RAG can ground responses in product descriptions and reviews if implemented.	Retrieval configuration and test examples
Storage	Local files; relational DB; Firestore	Paper identifies Firestore for profile/measurement data and ChromaDB for vector retrieval.	Actual schema and access rules

4.4 Detailed Design

4.4.1 Measurement and size-prediction flow

Input validation should verify required fields, plausible units, and supported categorical values. The backend then applies the same preprocessing used during training. The regression model returns the supported measurement estimates; the size-classification stage maps the predicted standard size to a brand-specific label. The response should include a clear indication of which outputs are estimates and which brand/size chart was used.

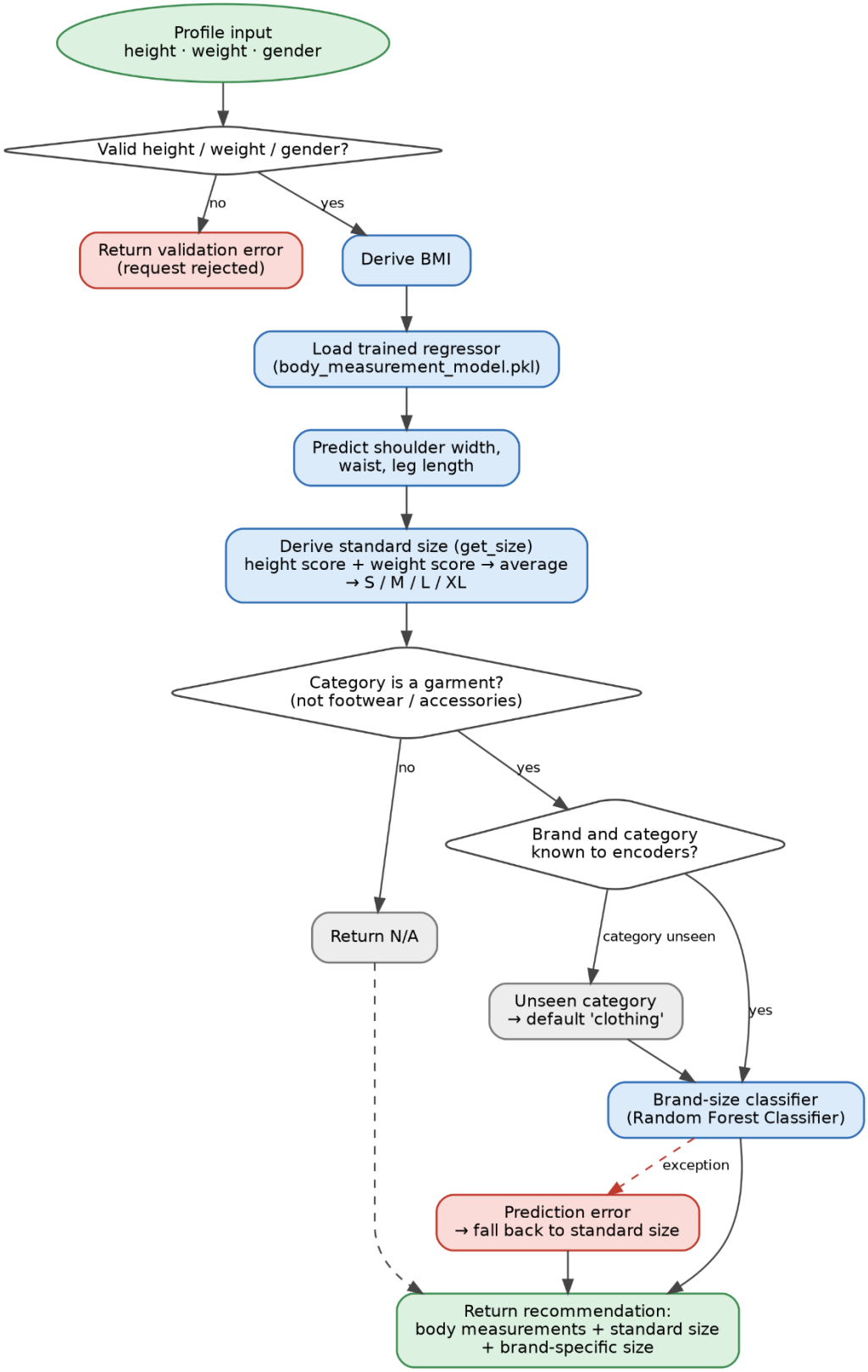


Figure 4.2: Measurement prediction and brand-size decision flow

4.4.2 Virtual try-on flow

The user selects or uploads a person image and garment image. The frontend checks required inputs and accepted formats; the backend sends the images to the configured virtual try-on service; the service returns a generated image or an error; the application displays the preview alongside the size recommendation. The thesis should document whether the image is retained, where it is stored, and how a user can remove it.

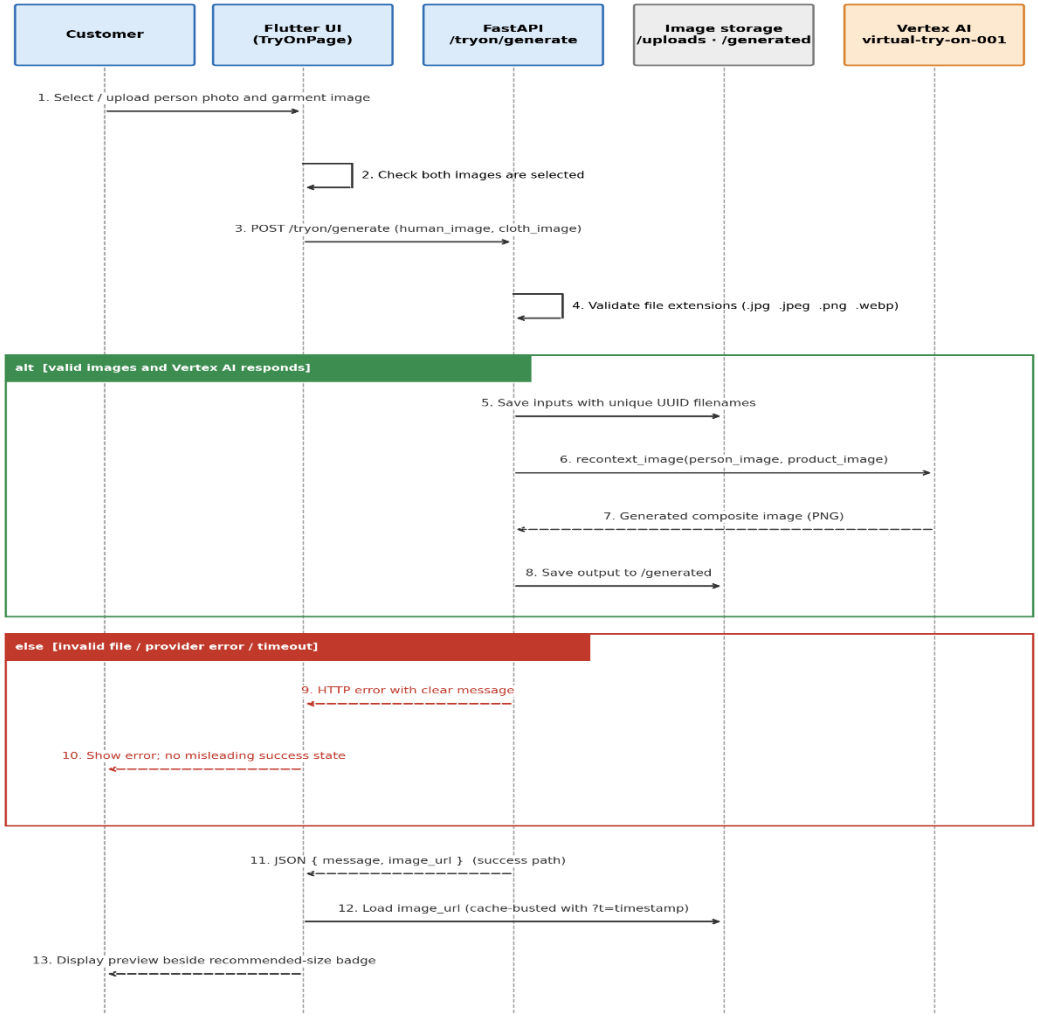


Figure 4.3: Virtual try-on request and response sequence

4.4.3 RAG stylist design (if included)

The RAG path starts with product text and reviews, processes them into document chunks, embeds them, and stores the vectors and metadata in ChromaDB. A customer query is embedded and used to retrieve relevant documents. The retrieved context is passed to the language model with an instruction to answer only from the supplied evidence. The application presents the answer and, where supported, the product facts or source items used.

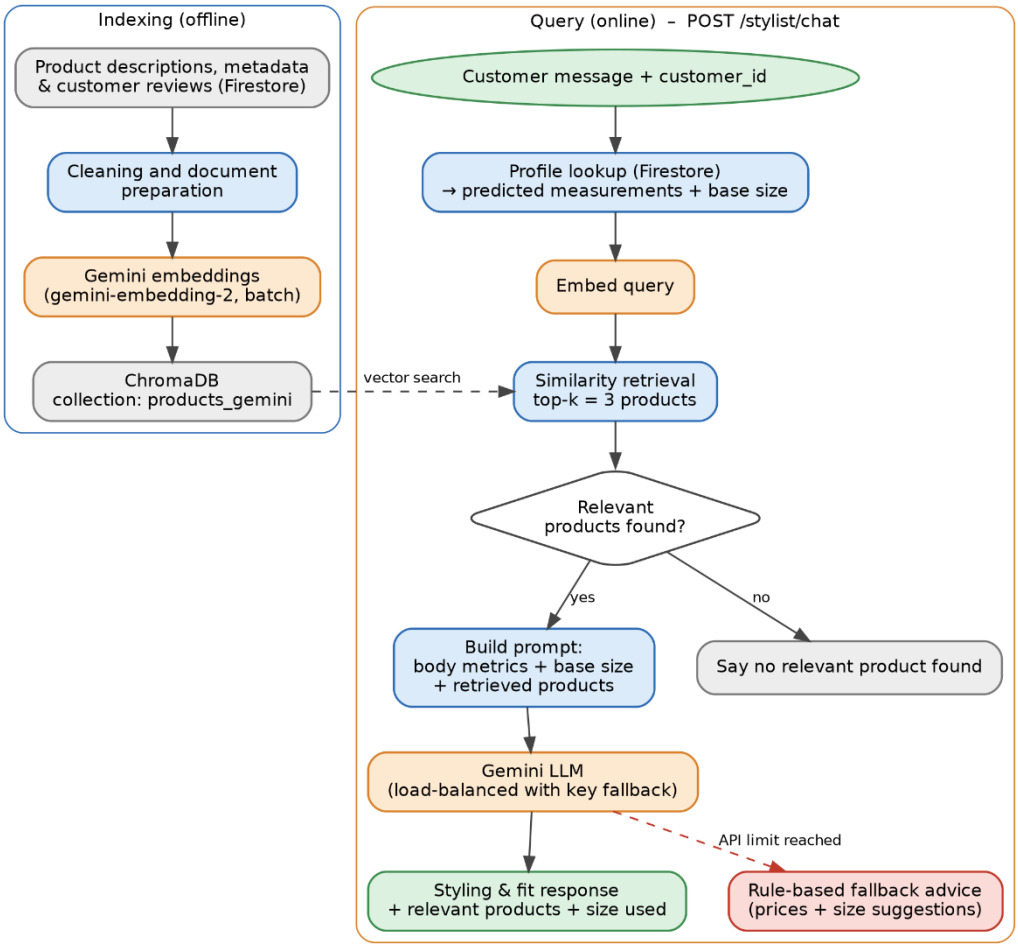


Figure 4.4: RAG-based virtual stylist architecture

4.4.4 Data model and API design

The final thesis should include only fields and endpoints confirmed from the implementation. At minimum, document the request/response schemas for measurement prediction, brand-size recommendation, virtual try-on, and (if present) stylist advice. Include data types, required fields, validation rules, success response, and representative error responses. Do not expose real user identifiers, secrets, or personal images in the thesis.

Table 4.4: API Services, Purposes, Inputs and Outputs

API / service	Purpose	Inputs to document	Outputs to document
Measurement prediction	Estimate supported body measurements	Height, weight, gender, derived BMI if used	Shoulder, waist, leg-length predictions and units
Size recommendation	Map to supported brand size	Brand, standard size or model inputs	Brand-specific size label and any confidence if actually returned
Virtual try-on	Generate garment visualization	Person image, garment image, required options	Image URL/file or provider response; errors
Stylist advice (optional)	Generate product-grounded guidance	User query, product identifier/context	Advice and retrieved evidence if exposed

4.5 Standards and Professional Practice

Use consistent units, input validation, secure secret management, version control, clear error handling, and accessible UI labels. Cite the applicable privacy legislation [16] and document image rights and third-party API conditions. The UN Sustainable Development Goals [17] can be discussed in Chapter 8, but claims of impact should remain proportionate to the prototype evidence.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1: Technologies and Tools Used in the System

Layer	Technology reported in the project paper	Confirm / add details
Frontend	Flutter	Version, platform, relevant packages
Backend	FastAPI / Python	Python version, routers, deployment
Measurement ML	Random Forest Regressor; GridSearchCV, 3-fold CV reported	Scikit-learn version, exact hyperparameters
Size classification	Random Forest Classifier; 100 trees reported	Features, encoding, class mapping
Virtual try-on	Google Cloud Vertex AI virtual-try-on-001 in paper; slide says Gemini-based	Exact endpoint/model ID and API parameters
Database	Firebase Firestore	Collections/documents, rules, retention
RAG (if implemented)	LangChain, ChromaDB, all-MiniLM-L6-v2	Versions, chunking, top-k, LLM model
Image hosting	GitHub Pages is listed in the paper's overall technology table	Confirm whether it is actually used by this component and whether it is appropriate for image handling
Version control	[Confirm]	Repository, branch, commit/tag

5.2 Component Implementation

5.2.1 Frontend

The customer interface is a Flutter application with a consistent glassmorphic design defined in a shared AppTheme, and all communication with the backend goes through a single ApiService HTTP client. The profile is created in CreateProfilePage, where the customer enters height (cm or inches) and weight (kg or lb). The helper functions _heightToCm and _weightToKg convert these to metric units before the request is sent. ProfilePage then shows the base size and an “AI Body Model” card,

which displays the predicted shoulder width, waist circumference and leg length in inches as labelled progress bars (Figure 6.4)

ProductDetailPage calls ApiService.predictBrandSize asynchronously when it loads and shows an “AI Recommended Size” card. If the brand-specific size differs from the standard size, the card explains that the size is adjusted for that brand’s fit pattern. If the two sizes match, it states that the standard size is the right fit for the brand. The page also has a “Try On” button that passes the product details to TryOnPage.

TryOnPage guides the customer through two steps. In Step 1 the customer picks a full-body photo from the gallery (pickHumanImage). In Step 2 the customer either selects a garment from the horizontally scrolling store catalogue or uploads a custom garment image (pickClothImage). While Vertex AI is processing, an animated pulsing button shows the loading state. When the result arrives, the screen shows a success message, a size-match badge with the recommended size (when recommendedSize is available), and the generated image. A timestamp query

parameter (?t=ms) is appended to the image URL to prevent a stale cached image from being displayed. A floating “Stylist” button opens the conversational stylist.

5.2.2 Backend and prediction services

The backend is a FastAPI application in Python. Its endpoints are listed in Table 5.2.

Table 5.2: Implemented API Endpoints of the Virtual Try-On and Size Recommendation Component

Endpoint	Method	Input	Output
/profile/create	POST	email, password, height, weight, gender	profile_id, recommended_size, body_measurements
/profile/update/{id}	PUT	height, weight, gender, password	profile_id, recommended_size, body_measurements
/profile/predict_brand_size	POST	standard_size, brand, category	brand_specific_size
/tryon/generate	POST	human_image, cloth_image (files)	message, image_url
/stylist/chat	POST	customer_id, message	response, relevant_products, used_size

Three saved scikit-learn artefacts are loaded by the service modules: body_measurement_model.pkl, brand_sizing_model.pkl and brand_sizing_encoders.pkl. The measurement model is a multi-output RandomForestRegressor with four input features (gender, height, weight, BMI) and three targets. The hyperparameters stored in the saved model are 500 trees, maximum depth 10, min_samples_split 5, min_samples_leaf 2 and random_state 42. To keep preprocessing the same as in training, the service encodes gender as Male =

1 and Female = 2 and calculates BMI with the same formula used in train_body_model.py.

The standard size comes from a rule-based function in profile_service.py. It scores the height (S below 165 cm, M below 175, L below 185, XL otherwise) and the weight (S below 60 kg, M below 75, L below 90, XL otherwise) separately. It then averages the two scores and rounds to a size from S to XL. The brand-specific classifier is a RandomForestClassifier with 100 trees and a seed of 42. It takes the standard size, brand and category, each encoded with a saved LabelEncoder. The encoders cover six brands and five categories.Error handling follows three fallback rules in brand_sizing_service.py: an unseen category defaults to “clothing”, footwear and accessories return “N/A”, and any prediction exception falls back to the customer’s standard size.

5.2.3 Virtual try-on integration

The virtual try-on uses Google Cloud Vertex AI’s virtual-try-on-001 model, called through the google-genai SDK in gemini_service.py with client.models.recontext_image. The module is named “gemini_service” for historical reasons, but the model that generates the image is the Vertex AI try-on model, not a Gemini language model. The customer’s photo is the person image and the garment is passed as a product image, and the request asks for one PNG with the safety filter set to BLOCK_LOW_AND_ABOVE.

The images are not base64-encoded by the app. The Flutter app sends them as multipart file uploads to POST /tryon/generate. The backend checks that the extensions are .jpg, .jpeg, .png or .webp and saves both files in an uploads/ folder under unique UUID filenames. The SDK then reads the images from those files. The generated composite is saved in a generated/ folder, which is served as static files,

and the endpoint returns a JSON response containing a message and the full public URL of the image. The app loads this URL to show the result.

5.2.4 RAG stylist implementation (if included)

The stylist is implemented in rag_service.py and exposed at POST /stylist/chat. Product records from the live store catalogue are embedded and stored in ChromaDB, which is saved locally in the backend's chroma_db folder as the collection products_gemini. Each entry carries metadata: product ID, name, brand, category, selling price, description, and customer reviews. A custom GeminiEmbeddings class uses the model gemini-embedding-2 and supports batch embedding.

When the customer sends a message, the backend runs the following steps: (1) it looks up the customer's height, weight and gender in Firestore; (2) it runs the measurement model and the standard-size rule to build a personal context block of shoulder width, waist, leg length and base size; (3) it retrieves the three most relevant products from ChromaDB with retrieve_relevant_products(message, top_k=3); (4) it builds a prompt from the body metrics, the base size and the retrieved items; and (5) it sends the prompt to a Gemini model through llm_manager, which can switch between several API keys. If the LLM limit is reached, the service generates a rule-based fallback answer containing the retrieved items' prices and size suggestions. The response includes the advice text, the relevant products and the size used.*Figure 1.1*

5.3 Integration with the Other Components

The supplied paper describes a shared FastAPI architecture and Firebase Firestore database across four system components. For this thesis, explain only the integration points that actually exist. Potential interfaces include shared product metadata and garment image references. Do not claim the virtual try-on component receives stock availability, marketing recommendations, or customer-location data unless the implementation demonstrates that exchange.

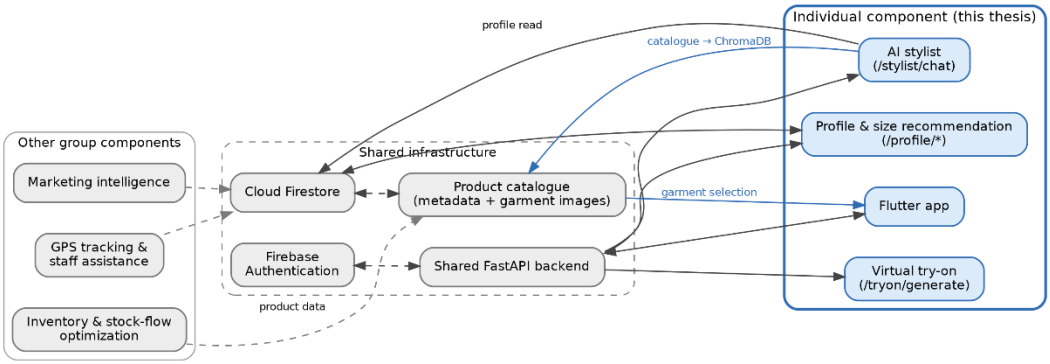


Figure 5.1: Integration with the wider smart retail system

5.4 Verification and Software Testing

Testing should cover valid inputs, missing inputs, invalid units, unsupported brands, malformed images, service timeout, external API failure, database failure, model output schema, and the end-to-end customer flow. Record actual expected/actual outcomes and evidence. Do not report a pass rate until it is calculated from the completed test table.

Table 5.3: Functional Test Cases and Results

Test ID	Scenario	Expected result	Actual result	Status / evidence
T01	Valid profile attributes submitted	Prediction response contains all required measurement fields and units	[Insert]	[Pass/Fail + screenshot/log]
T02	Missing required profile field	Clear validation message; no invalid prediction	[Insert]	[Insert]
T03	Invalid height/weight units or range	Request rejected or corrected according to documented rules	[Insert]	[Insert]
T04	Supported brand and valid size input	Brand-specific size response returned	[Insert]	[Insert]
T05	Unsupported brand	Clear unsupported-brand message or documented fallback	[Insert]	[Insert]
T06	Valid person and garment images	Try-on result returned and displayed	[Inser	[Insert]
T07	Invalid image format	Clear validation error	[Insert]	[Insert]
T08	Virtual try-on provider timeout/failure	Graceful error; no misleading success state	[Insert]	[Insert]
T09	Database unavailable (if relevant)	Graceful failure and no data corruption	[Insert]	[Insert]
T10	End-to-end try-on and size workflow	Size recommendation and generated preview are both displayed	[Insert]	[Insert]
T11	RAG question with relevant product data (if in scope)	Response grounded in retrieved product information	[Insert]	[Insert]
T12	RAG question with no relevant evidence (if in scope)	System abstains or clearly indicates insufficient information	[Insert]	[Insert]

5.5 Deployment and Operation

State whether the frontend and backend run locally, on a cloud service, or through another host. Describe environment variables, secret storage, model artefacts, database configuration, external API dependencies, and startup steps. Do not include secrets. If the system was not deployed to real retail users, state that it was tested as a prototype.

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 Body-measurement regression performance

The supplied research paper reports the following R^2 values and MAE values for the measurement model. The paper does not provide the dataset size, train/test split, test period, or full RMSE results in the visible table, so those details must be confirmed before the table is treated as complete.

Table 6.1: Body Measurement Prediction Model Performance

Target	Reported MAE	Reported R^2	RMSE	Unit / test-set details
Shoulder width	1.12	0.914	[Not reported in supplied paper]	Confirm cm and held-out test setup
Waist circumference	1.45	0.897	[Not reported in supplied paper]	Confirm cm and held-out test setup
Leg length	1.28	0.921	[Not reported in supplied paper]	Confirm cm and held-out test setup

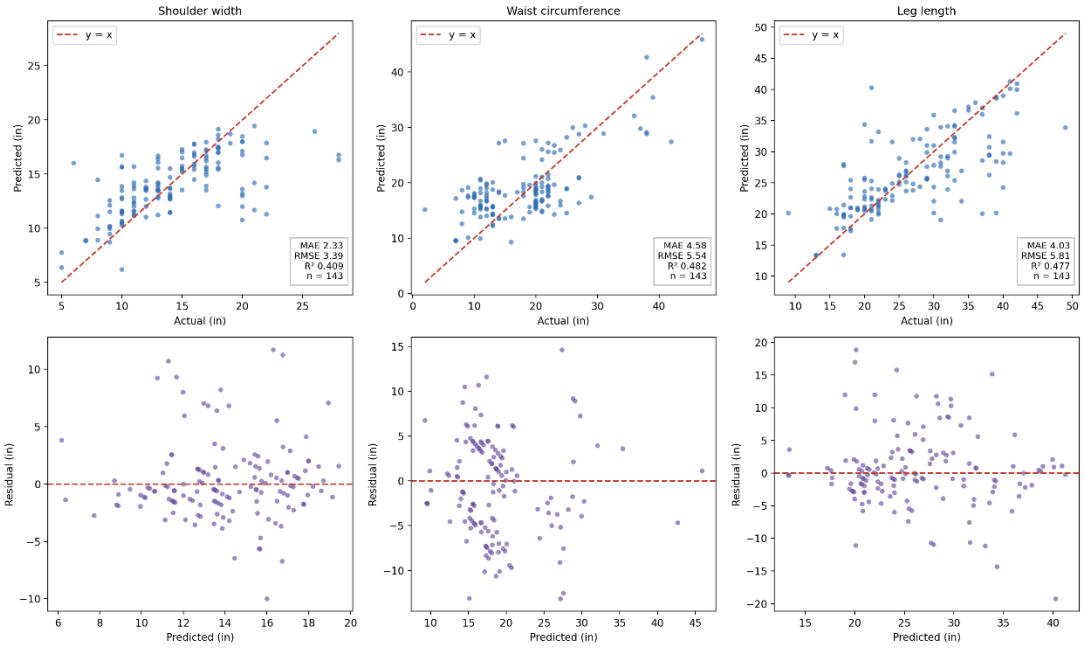


Figure 6.1: Body-measurement regression performance

6.1.2 Brand-specific size classification

The supplied paper reports an overall classification accuracy of 88.4%, weighted precision of 0.879, weighted recall of 0.884, and weighted F1-score of 0.881 for brand-specific size classification. Confirm the class definitions, brand coverage, test-set size, and split procedure.

Table 6.2: Brand-Specific Size Classification Model Performance

Metric	Reported result	Required clarification
Accuracy	88.4%	Test sample count and class labels
Weighted precision	0.879	Per-class scores and class support
Weighted recall	0.884	Per-class scores and class support
Weighted F1-score	0.881	Per-class scores and class support

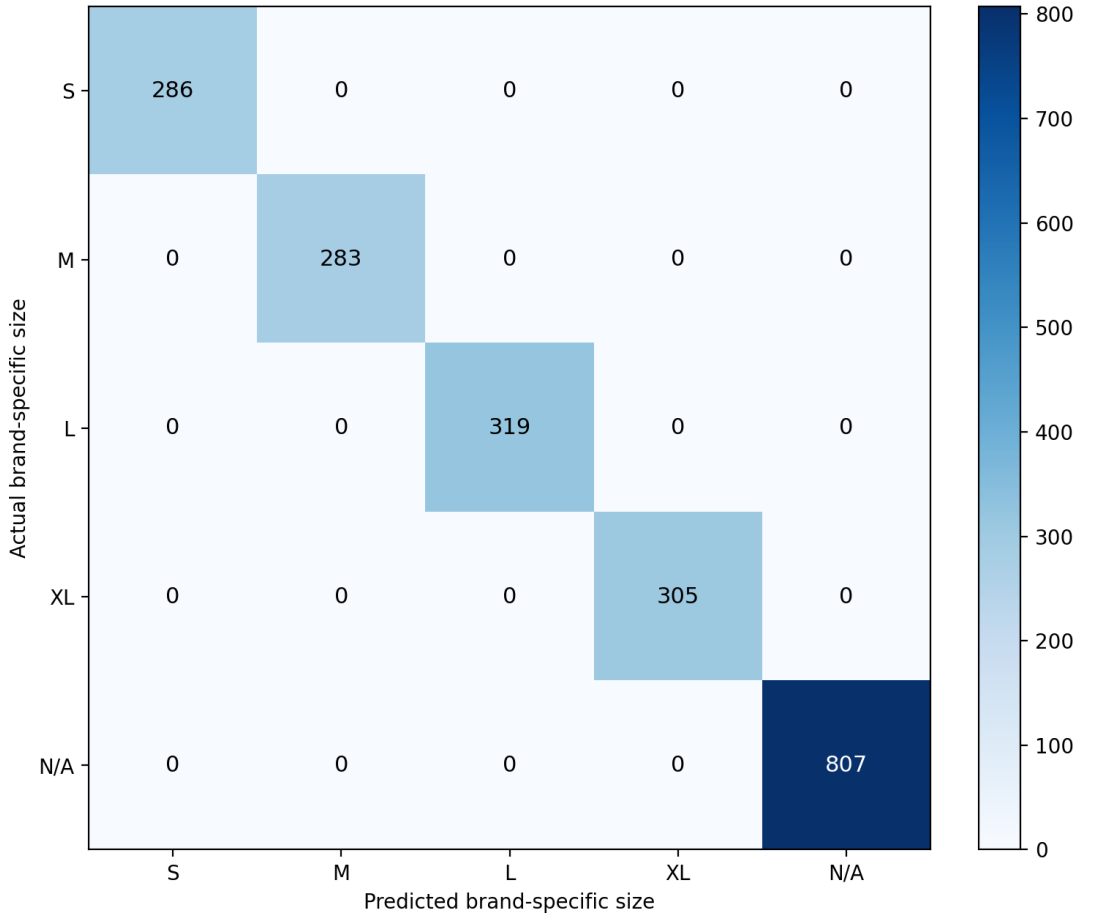


Figure 6.2: Brand-specific size classification confusion matrix.

6.1.3 Virtual try-on quality and technical performance

The supplied paper describes the virtual try-on approach but does not provide a quantitative visual-quality metric, response-time distribution, failure rate, or human-evaluation result for this component. This section must report only tests actually conducted. At minimum, provide a representative set of input/output examples and a structured account of failures such as pose changes, occlusion, garment-detail changes, or unsupported images. If no formal quality study was performed, state this limitation.

Table 6.3: Virtual Try-On Testing and Visual Quality Evaluation

Test case	Input description	Observed output	Quality issue / success criterion	Evidence
VTO-01	[Person image and garment category]	[Describe actual output]	[Describe criteria]	[Screenshot/file reference]
VTO-02	[Different supported garment or pose]	[Describe actual output]	[Describe criteria]	[Screenshot/file reference]
VTO-03	[Edge case / difficult image]	[Describe actual output]	[Describe criteria]	[Screenshot/file reference]

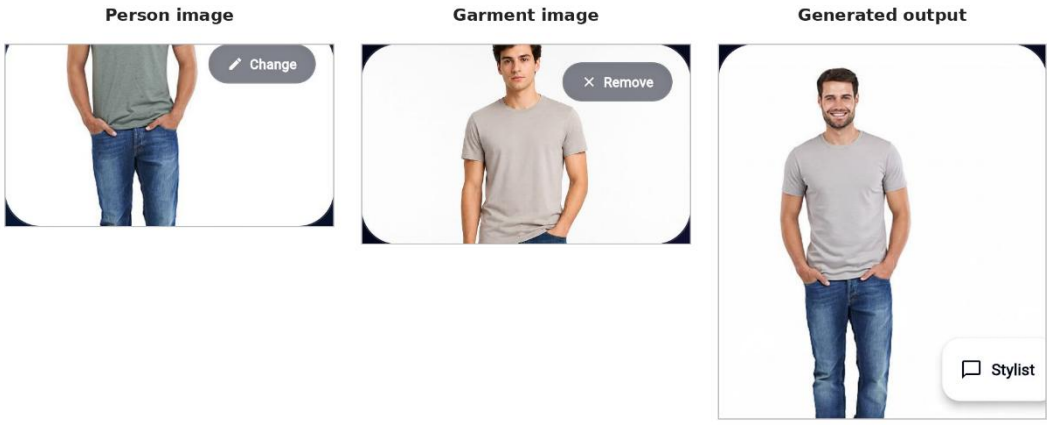


Figure 6.3: Virtual try-on input and generated output examples

6.1.4 RAG stylist evaluation (if included)

The paper describes the RAG pipeline but does not provide quantitative retrieval or groundedness results. If the feature was evaluated, report the test questions, expected evidence, retrieved documents, answer quality rubric, and unsupported-claim findings. If not evaluated, describe the implemented pipeline and mark formal RAG evaluation as future work.

6.2 Product and Technical Results

Report completed screens, supported workflows, API behaviour, response times, failure handling, and integration status. Keep model accuracy separate from virtual try-on image quality and application functionality.

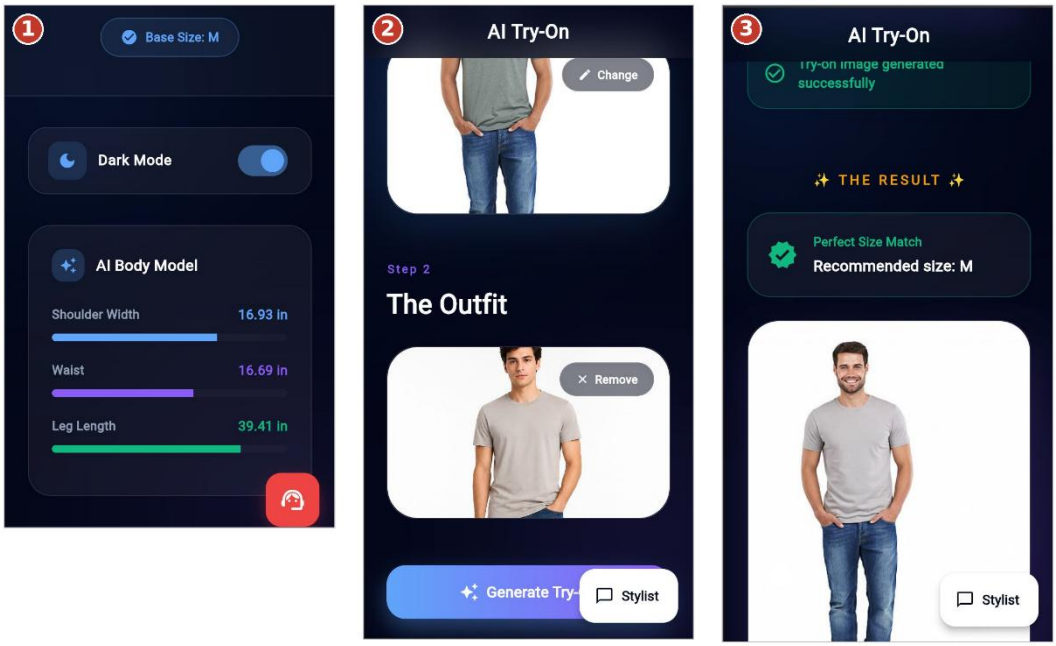


Figure 6.4: End-to-end application workflow evidence

6.3 User or Stakeholder Validation

No user study, participant count, questionnaire, or expert review protocol is established by the supplied paper. If validation was conducted, document participants, recruitment, consent, tasks, questions, rating scale, analysis, and

limitations. If no participants were involved, state that evaluation was limited to model metrics and technical tests; do not invent satisfaction scores.

6.4 Integration Results

Report which shared services and product data interfaces were successfully tested. Include evidence for data exchange only where confirmed. If the wider system integration was not completed, describe the planned interface and mark it as not yet validated.

6.5 Summary of Findings

The available paper indicates promising preliminary results for the reported measurement and brand-size models. However, a complete validation requires details of the dataset, held-out test procedure, model configurations, virtual try-on evaluation, and end-to-end testing. The final summary should answer each research question using the actual evidence and should not treat an R^2 or classification accuracy score as proof that generated images depict physical fit accurately.

7 DISCUSSION

7.1 Interpretation of Findings

Discuss measurement-regression errors in their physical units and explain what they mean for a size decision. A high R^2 does not mean that every individual prediction is close to the true measurement. For size classification, discuss class imbalance, confusions between adjacent sizes, and whether weighted metrics hide poor performance for less frequent classes. Discuss generated-image quality independently from the numeric models.

7.2 Comparison with Literature, Baselines and Existing Solutions

Compare the component with related virtual try-on and size recommendation studies reviewed in Chapter 2. Use the same target definitions and evaluation protocol where possible. If the project did not run a baseline or reproduce a published model, describe the comparison as qualitative rather than experimental.

Table 7.1: Comparison of the Proposed System with Existing Systems

Dimension	**Proposed component**	**Comparison evidence required**
Input requirements	Profile attributes and person/garment images as implemented	Comparable systems' documented inputs
Body measurement	Random Forest regression	Same targets, units, test design and metrics
Size recommendation	Brand-specific classifier	Brand coverage, size classes, class-wise performance
Visual try-on	Managed AI try-on service	Image-quality rubric and representative outputs
Grounded stylist (optional)	RAG using product metadata/reviews	Retrieval relevance and unsupported-claim test

7.3 Research Contribution

The research contribution is the design and evaluation of an integrated workflow combining body-measurement prediction, brand-specific size recommendation, and AI-generated garment visualization. If the RAG stylist is part of the completed scope, it adds a product-grounded natural-language advice path. State the contribution in proportion to the work actually implemented and evaluated.

7.4 Product Contribution

The product contribution is a customer-facing digital fitting-room workflow intended to make online clothing selection more informative. The system presents a visual preview alongside a separate size recommendation, helping users consider appearance and size guidance together. The system does not guarantee actual garment fit.

7.5 Limitations and Threats to Validity

• Dataset representativeness: training data may not cover the full range of body shapes, age groups, garment categories, and brands.

• Measurement validity: height, weight, gender, and BMI may not capture enough information to estimate all body dimensions accurately.

• Label quality: measurement or size labels may contain noise, inconsistent units, or brand-specific inconsistencies.

• Evaluation leakage: preprocessing, feature selection, or model tuning may leak information if train/test separation is not handled correctly.

• Visual generation: output images can contain artefacts, alter garment details, or fail on difficult poses and occlusions.

• External-service dependency: model changes, quotas, latency, and outages may affect repeatability and operation.

• RAG reliability: advice depends on the completeness and freshness of product metadata and retrieval quality.

• Limited user validation: if no user study was performed, usability and purchase-impact claims remain unverified.

7.6 Implications

For customers, the component may reduce uncertainty by providing both a preview and a size suggestion. For retailers, it may offer a more engaging product-discovery experience. However, the prototype's effects on returns, conversion rates, customer satisfaction, and commercial performance require real-world evaluation and should not be claimed from model metrics alone.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The primary users are online apparel shoppers who need help visualizing clothing and choosing a size. Secondary stakeholders may include fashion retailers seeking to improve digital product pages and customer engagement. The value proposition is a combined visual preview and size recommendation within one workflow, reducing reliance on separate tools. These benefits are proposed rather than commercially proven unless user or retailer evidence is available.

8.2 Market and Competitive Position

The component can be positioned as a digital fitting-room feature for fashion e-commerce. A final market analysis should identify current virtual try-on and sizing products, their supported categories, integration models, pricing where published, privacy terms, and limitations. Do not use unsourced market-size figures or make unsupported competitor comparisons.

8.3 Business or Sustainability Model

Potential delivery models include a retailer subscription, per-request API usage, integration as an e-commerce plug-in, or a custom enterprise deployment. These are possible options, not an existing business model. A feasibility analysis should estimate cloud inference costs, image storage, maintenance, support, and expected usage before proposing pricing.

Table 8.1: Cost and Value Analysis of the Proposed System

Cost / value factor	What to estimate	Evidence required
Virtual try-on inference	Cost per generated image and expected monthly volume	Provider pricing and usage assumptions
Backend and database	Hosting, storage, bandwidth, monitoring	Actual deployment plan and provider pricing
Model maintenance	Retraining, brand-size updates, testing	Update frequency and staff effort assumptions
Retailer value	Potential reduction in size uncertainty/returns, engagement	Pilot study or retailer evidence; do not assume savings

8.4 Scalability, Intellectual Property and Compliance

Scalability depends on image-generation latency, API quotas, concurrency, image size, model-serving costs, database access patterns, and the number of supported products and brands. Intellectual-property review should cover ownership/licensing of product images, training data, generated outputs, software dependencies, and third-party model terms. Privacy and compliance discussion should include the Sri Lankan Personal Data Protection Act [16] and any other applicable rules identified by the institution or project supervisor.

8.5 Contribution to the Sustainable Development Goals

The UN 2030 Agenda [17] provides a framework for discussing possible sustainability relevance. The component may relate to responsible consumption and production (SDG 12) if future evidence shows that improved fit decisions reduce avoidable returns or waste. It may also support digital innovation. These are potential contributions, not measured impacts; claims should be supported by a pilot or lifecycle analysis.

8.6 Risk Analysis and Mitigation

Table 8.2: Risk Assessment and Mitigation Strategies

Risk	Potential impact	Possible mitigation
Incorrect size prediction	Customer dissatisfaction or return	Clear uncertainty wording, broader training data, per-class testing
Misleading generated image	Customer expects an unrealistic appearance/fit	Label image as simulation; evaluate garment-detail preservation
Sensitive image exposure	Privacy harm	Minimize retention, access controls, secure transfer, deletion process
External API outage or cost increase	Service unavailable or uneconomic	Timeouts, quota monitoring, graceful fallback, cost limits
Unsupported brand or garment	Unreliable recommendation	Show coverage limits and avoid extrapolating beyond supported classes

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis concerns the design of a Virtual Try-On and Size Recommendation System for online apparel shopping. The component combines a machine-learning approach to estimate selected body measurements, brand-specific size classification, and AI-based garment visualization. The supplied research paper also describes a RAG stylist that retrieves product descriptions, metadata, and customer reviews to generate product-grounded advice. The final conclusion must be revised after the dataset, implementation, and evaluation evidence have been confirmed.

9.2 Achievement of Objectives

Table 9.1: Research Objectives and Achievement Status

Objective	Evidence required	Status
Review relevant literature	Completed literature review and reference list	[Confirm]
Prepare model data	Dataset description, preprocessing code, record counts	[Confirm]
Develop measurement model	Training configuration and saved model	[Confirm]
Develop brand-size classifier	Training code, class labels, test metrics	[Confirm]
Integrate virtual try-on	API integration and screenshots/logs	[Confirm]
Evaluate the component	Held-out metrics, visual evaluation, test table	[Partially evidenced in paper; confirm]
Test integrated workflow	Completed test-case table and evidence	[Confirm]

9.3 Individual Research and Product Contributions

[AUTHOR INPUT REQUIRED] Summarize the work you personally designed, implemented, tested, and documented. Do not assign shared infrastructure or other group members' work to yourself. Explain the contribution beyond merely calling an external API, such as workflow design, data preparation, model integration, size mapping, validation, or responsible presentation of outputs, where these are genuine.

9.4 Future Directions

• Evaluate with larger, more representative body-measurement and brand-size datasets.

• Add more brands and garment categories with verified size charts and garment measurements.

• Evaluate end-to-end size accuracy using the predicted measurements rather than ground-truth intermediate values.

• Conduct a structured human evaluation of visual fidelity, garment-detail preservation, and perceived usefulness.

• Improve support for pose variation, occlusion, lighting differences, and image-quality problems.

• Add confidence/uncertainty indicators and a safe fallback when the model cannot make a reliable recommendation.

• Evaluate RAG retrieval relevance and answer groundedness with a defined benchmark if the stylist is included.

• Conduct a privacy, security, and retention review before any real customer deployment.

• Run a retailer or customer pilot to test usability and potential effects on purchase confidence and returns.

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GLOSSARY

Term	Definition
AI	Artificial Intelligence; methods that perform tasks commonly associated with human intelligence.
BMI	Body Mass Index, calculated as mass in kilograms divided by height in metres squared.
ChromaDB	Vector database used in the described RAG pipeline to store and retrieve embeddings.
MAE	Mean Absolute Error; average absolute difference between predicted and true values.
ML	Machine Learning; algorithms that learn patterns from data.
R^2	Coefficient of determination, a measure of regression fit relative to a baseline.
RAG	Retrieval-Augmented Generation; retrieval of external knowledge to condition generated responses.
RMSE	Root Mean Squared Error; square root of the average squared prediction error.
VTO	Virtual Try-On; image generation that visualizes a garment on a person image.
XAI	Explainable Artificial Intelligence; methods for making model outputs more interpretable.

APPENDICES

Appendix A – Extended Results and Research Artefacts

• A.1 Dataset schema, data dictionary, provenance, licensing, and preprocessing record.

• A.2 Full regression results by target, including MAE, RMSE, R^2, sample counts, and residual summaries.

• A.3 Full classification report and confusion matrix by size class/brand.

• A.4 Virtual try-on input/output examples and documented failure cases, with permission and personal data removed.

• A.5 RAG test questions, retrieved context, generated answers, and grounding evaluation, if applicable.

• A.6 API request/response examples with secrets and personal data removed.

- A.7 Full software test cases, expected/actual outcomes, pass/fail status, and screenshot/log references.
- A.8 Model hyperparameters, software versions, environment details, and reproducibility notes.
- A.9 User study materials and consent forms, only if user testing was conducted.

Appendix B - Academic Integrity and AI Use Disclosure

This appendix records how generative AI tools were used in producing this thesis, in line with the module policy that AI may assist the work but cannot become the owner of the work. It is a structure for me to complete and verify. I remain responsible for the accuracy, originality, citations, code, data, analyses and conclusions of the thesis.

Two kinds of AI use should be kept apart. The first is AI used as a component of the product, namely Gemini for explanations and a conversational model for the chatbot, which are described in Chapters 4 and 5. The second is AI used as an assistant while producing the thesis, code or figures, which is what this disclosure covers.

Table B.1: AI use disclosure

AI tool and version	Purpose	Representative workflow	Human verification performed	Sections or artefacts affected
Claude (Anthropic) [CONFIRM tool, version and dates]	Drafting the first version of the thesis text from my template, formatting guidelines and project information	I supplied the template, the formatting standards and my project facts and asked for a complete first draft with placeholders for missing evidence	[INFORMATION REQUIRED: describe what you checked and edited, for example results against your records, every citation against its source, and technical descriptions against your code]	[INFORMATION REQUIRED: chapters and sections drafted with AI help]
[Other tool, if any]	[For example debugging ideas, test-case suggestions, language improvement]	[INFORMATION REQUIRED]	[INFORMATION REQUIRED]	[INFORMATION REQUIRED]

B.1 Declaration statement

[INFORMATION REQUIRED: a short statement in your own words, for example: I used the tools listed above for the purposes stated. I reviewed and edited all AI-assisted text, checked the technical content against my implementation and results, verified each citation, and take full responsibility for the content of this thesis.]

ITEMS THE STUDENT MUST VERIFY OR ADD BEFORE SUBMISSION

Working page. Delete before submission. Every yellow-highlighted item in the thesis is also a placeholder.

A. Personal and administrative details

- Full student name, group ID, supervisor and co-supervisor names, official degree title, specialisation (if any), department, and submission month and year (title page, declaration, acknowledgements).
- Confirm the thesis title and subtitle wording on the title page.
- Sign and date the declaration, and obtain supervisor signatures.
- Names and IDs of the other group members in Table 1.1, and confirm that the Smart Mirror is component 3.
- Acknowledgements: add only assistance actually received, plus any funding statement.

B. Data and machine-learning facts

- How the synthetic dataset was generated and enhanced (effects encoded, noise, seed), and how the 476,000 records relate to the full product-store-day grid.
- Handling of missing values and non-selling days.
- Train and test split (ideally chronological), hyperparameters, whether cross-validation was used, and size and period of each evaluation set.
- Confirmation that rolling_mean_7 and the lag features contain no leakage of the predicted day.
- Features, training procedure and metrics of the monthly and quarterly models. Quarterly metrics are missing.
- Mean or median daily and monthly sales in the test set, so MAE can be interpreted, and any additional metric you choose to measure (marked [Metric to be added after final evaluation]).
- Optional but strongly recommended: a naive or moving-average baseline, so the reported metrics have a reference point.
- How multi-day forecasts are produced from lag-based features, and whether accuracy at longer horizons was examined.

C. Design and implementation facts

- Exact rules and thresholds for SHORTAGE, HEALTHY and OVERSTOCK, low stock, reorder prediction and slow-moving stock.
- The ranking rule for source stores, the logistics-cost formula, the risk and confidence rules, and the handling of partial coverage, ties and missing distances.

- Firestore collection names for forecasts, store distances and the explanation cache, the cache key, and when cached explanations are invalidated.
- Whether execution uses a Firestore transaction or batch write.
- Gemini model version, prompt structure, and how explanations were checked against the supplied facts.
- How chatbot intent detection is implemented and how answers were validated.
- API endpoint names for Table 5.2, and whether Swagger UI was used in testing.
- Security details: key storage, Firestore security rules, and role checks for approval and execution (also NFR9). Confirm that no customer personal data reaches the AI services.
- Software versions, hosting and deployment arrangement (Section 5.5), and any response-time measurements (NFR10). No performance figures have been stated.
- Confirm which parts of the Flutter interface you built, and that the frontend is Flutter Web only. If an earlier prototype used another framework, say so.

D. Testing evidence

- Final test-case table for TC01 to TC16 with inputs, expected and actual results, pass or fail and evidence (Section 5.4 and Appendix A.2). No pass percentages appear in this draft, and none should be added unless calculated from that table.
- Defects found during testing and how they were fixed, if you kept a log.

E. Validation and integration evidence

- Stakeholder validation (Section 6.3): participants, procedure, instruments, real results, supervisor feedback, and PP3 user-feedback videos or images. Nothing has been invented here.
- Integration evidence (Section 6.4): a screenshot of the Smart Mirror using an image URL, and confirmation from that member. Describe the marketing-component integration only if it exists.
- Explanation quality (Section 6.2): how many explanations you inspected and whether any contained unsupported statements.

F. Literature and citations

- Section 2.1: databases, search strings, date of last search, and inclusion criteria.
- Every marker: fashion-retail demand characteristics, weather or festival effects on apparel demand, inventory definitions (textbook), random forest versus boosting, named competing products, conversational decision support, market size and inventory-loss figures, and Firestore pricing and quotas.
- Verify the M5 statement in Section 2.3.1 against the paper. Its bibliographic details were confirmed in a search, but the specific claim you make about the results still needs checking.
- The following entries were confirmed through searches but you should check the final details against the publisher pages before submission: Naderi et al. (article number),

Hyndman and Athanasopoulos (edition and URL), Grinsztajn et al. (volume and track), the Sri Lanka PDPA citation (official gazette or Parliament source and current commencement status), and the UN 2030 Agenda entry (A/RES/70/1).

- The reference list contains only sources cited in the text. If you delete a sentence, delete its reference if no other sentence cites it, and add more peer-reviewed sources from 2020 to 2026 where you can, especially for conversational decision support and LLM-based explanation.

G. Figures and screenshots to create

- All figure placeholders are listed below with their numbers. Replace each placeholder box with the image, keep the caption, and delete the specification text. Do not leave a placeholder in the final version.

H. Word and formatting tasks

- Press Ctrl+A then F9 in Word to update the table of contents, list of figures, list of tables and page numbers, and accept the update prompt on opening.
- Delete or resolve every highlighted placeholder. Yellow highlighting marks every item that needs your input.
- Check the abstract length (300 words maximum) after any edits, and its single spacing.
- Check fonts and spacing (Times New Roman, 12 pt, 1.5 spacing, margins 40, 25, 25 and 40 mm) after pasting into the official template. The draft uses these settings, but the template's styles may differ.
- Spelling: the draft mixes some British forms (for example behaviour, summarised) with the -ize spelling used in the component title (optimization). Choose one convention, apply it consistently, and keep the template's own heading spelling (Commercialisation) if the department requires it.
- Remove this final page before submission.
- Draft thesis deadline: 11 October 2026. Final submission: 28 October 2026. PP3 evidence: proof folder, logbook, MS Planner, Teams and GitHub history.